Example Code: Symbols and Characters()

So far, we have only displayed simple characters on the LCD by using what is easily entered from our keyboard. To make more interesting symbols on the display, the HD44780 driver has a number of special symbols, like ¢, °, Ω, or π, which can be accessed using a code that tells the display what symbol to show. These character displays also have the capability of displaying up to eight custom characters of our own design. The source code in Listing 9-2 provides an example of using both the built-in symbols as well as some new ones that we will create ourselves.

Listing 9-2. *Symbols and Characters Source Code*

```
#include <LiquidCrystal.h>
LiquidCrystal lcd(5, 6, 7, 8, 9, 10);

byte degree = B11011111;
byte cents = B11101100;
byte sqRoot = B11101000;
byte divide = B11111101;
byte pi = B11110111;
byte omega = B11110100;
byte rgtArrow = B01111110;
byte lftArrow = B01111111;

byte symbols[] = {degree, cents, sqRoot, divide, pi, omega, rgtArrow, lftArrow};

byte smiley[] = {B00000,B00000,B01010,B00000,B10001,B01110,B00000,B00000};
byte skull[] = {B00000,B01110,B10101,B11011,B01110,B01110,B00000,B00000};
byte bell[] = {B00000,B00100,B01110,B01110,B01110,B11111,B00100,B00000};
byte note[] = {B00000,B00100,B00110,B00101,B00101,B01100,B01100,B00000};
byte heart[] = {B00000,B00000,B01010,B11111,B11111,B01110,B00100,B00000};
byte fish[] = {B10000,B00000,B01000,B10000,B01101,B11110,B01101,B00000};
byte lock[] = {B00000,B01110,B10001,B10001,B11111,B11011,B11011,B11111};
byte unlock[] = {B00000,B01110,B10001,B10000,B11111,B11011,B11011,B11111};

byte* characters[] = {smiley, skull, bell, note, heart, fish, lock, unlock};

void setup() {
  lcd.begin(16, 2);

  for (int i=0; i<8; i++) {
    lcd.setCursor((i*2),0);
    lcd.write(symbols[i]);

    lcd.createChar(i, ((byte*)characters[i]));
    lcd.setCursor((i*2),1);
    lcd.write(i);
  }
}

void loop() {}
```